

Handoff: top-K Lyapunov measures, non-normal amplification, intermittency, stimulation

Written 2026-09-13 13:50 on R5611351, Claude Code session 90c5825b-19c1-4546-90a4-703d562404ba, repo FractionalReservoir, branch main, HEAD 3a64568 ("Drop the config-level model override; the top-K cap is 60 in analysis_run_config"). Working tree clean. Written because the session is near its context limit; the next session should be able to resume from this file plus the commit messages of 5a9d409...3a64568.

1. What was built in this session (2026-09-12 → 13)

All committed on main. Read the commit messages: they record the numbers and the wrong turns.

1.1 Estimators and model features (earlier in the session)

- **lya_method = 'topk'** (src/model/lyapunov/lyapunov_topk.m): the K largest Lyapunov exponents by the discrete QR method on the stored trajectory; Heun on the fiducial grid; economy QR every lya_dt; returns LE_spectrum, local_LE_spectrum_t, finite_LE_spectrum_t, t_lya, n_positive, h_KS(_bits), D_KY, D_KY_resolved, cond_max, orth_defect_max, Q_final. Verified vs the ode45 QR, Benettin and Liouville (scripts/tests/test_lyapunov_topk.m).
- **Matrix-free Jacobian product** SRNNCellTypePairs.jacobian_times(S, Y, params) (= compute_Jacobian_fast * Y to 1e-16; test_jacobian_times). Any change to compute_Jacobian_fast or dynamics_fast MUST be mirrored. Top-10 on the 4000-state network: 139 → 8 s.
- **Per-neuron SFA ladders** tau_a_spread / tau_a_seed / read-only tau_a_matrix (log-normal at both ladder ends, geometric between; preset...tauSpread0p05...; test_SRNNCellTypePairs_tau_a_spread). Measured: does NOT speed Benettin or top-K warm-up (docs/notes/Lyapunov_estimation_methods.md §2.4).
- **Retry on an unresolved D_KY**: lya_K_auto / lya_K_max on the class; doubles K on the stored trajectory; nests exactly (test_lya_K_auto).
- **SRNNCellTypePairs.lya_summary()** — the one definition of the per-run scalars: LLE, lambda_2, lambda_gap, n_positive, h_KS_bits, D_KY, D_KY_resolved, K_used, retries, n_positive_at_K, cond_max, orth_defect_max, lambda_1_drift, frac_local_positive, p95_finite_0p2s, mean_positive_excursion_s, lead_frac_x/_sfa/_std/_stf, plus the leading direction's local-rate series. Statics lya_summary_fields, transient_divergence(local, dt, window_s), leading_vector_blocks.

1.2 Sweeps, registry, figures, stages (this part of the session)

- **src/presets/analysis_run_config.m** pack(): every sweep and mode now uses lya_method 'topk', lya_K 15, lya_K_auto true, lya_K_max 60, lya_dt 0.05. Benettin comparison: a config sets cfg.model.lyapunov_method = 'benettin' after calling it (same seeds, same networks). NO config-level model_overrides layer (added and removed again on TR's request — 6eb7c69/3a64568).
- **ParamSpaceAnalysis2.run_single_job** copies every lya_summary scalar into each result (nan_lya_fields for failures) plus local_rate_lead / t_lya_lead.
- **src/presets/sweep_metrics.m** — the registry of measures the figures plot (key, field, label, stem, ranges, zero line, in_sheets). Sheets now exist for lle, r, hks, dky: fig_sensitivity_medians (Fig_Sensitivity_<stem>_medians), fig_sensitivity_analysis_allStd (Fig_Sensitivity_<stem>_core/_mu), fig_param_space_allStd (Fig_ParamSpace_<stem>), fig_EI_param_space / fig_EI_weights_param_space (Fig_EI_[Weights_]ParamSpace_<stem>, shared layout in src/figures/helpers/ei_metric_sheet.m). fig_sfa_EOC_allStd gained Fig_SFA_EOC_blocks (leading-vector SFA/STD/x fractions vs $\tau_{\max}$: 80-85% SFA across 1-30 s).
- **write_run_parameters_md**: a **Lyapunov quality** table per sweep and condition (share D_KY resolved, share n_+ at K, K used, worst conditioning, median $|\lambda_1|$ drift, orthogonality defect).
- **src/analysis/run_eig_heatmap.m** now runs top-K and records, per sampled state, the numerical abscissa $\omega = \max \text{eig}((J_{xx} + J_{xx}^T)/2)$ and spectral abscissa $\alpha = \max \text{Re eig}(J_{xx})$ of the **dendritic block** (num_abscissa_by_cond, spec_abscissa_by_cond, J_times_by_cond, lya_by_cond). Figure: src/figures/fig_transient_amplification.m.
- **src/analysis/run_lyapunov_spectrum.m** + **src/figures/fig_lyapunov_spectrum.m**: per regime, seeds, noise on AND off, top-K at K = 50/200/300 by mode, lya_K_max 2K; n_override for tests.

Registered in `run_all_paper_analyses` (stage `lyapunov_spectrum`) and in `paper_config`, `single_multi_TS_independent_config`, `_med_config`, `topk_smoke_fast_config`, `topk_med_config`; `test_run_modes` lists it.

- Tests added: `test_lya_K_auto`, `test_sweep_metrics`, `test_lyapunov_spectrum_stage` (+ `test_jacobian_times`, `test_SRNNCellTypePairs_tau_a_spread` earlier). Baseline suite that passes today (2026-09-12): `test_SRNNCellTypePairs`, `S_c_heterogeneity`, `test_lyapunov_topk`, `test_preset_golden`, `test_preset_conditions`, `test_run_modes`, `test_sensitivity_refactor`, `test_psa_model_class`, `test_psa_loaders`, `test_psa_validate_default`, `test_psa_saveload`, `test_figure_report`, `test_resolve_data_file`, `test_c_over_K`, `test_n_a_dependent`, `test_pairs_single_celltype`, `test_esn_route_redundancy`. Many other tests in `scripts/tests/` are stale (SRNNModel2-era); do not rely on them.
- Configs/runs: `scripts/paper/topk_smoke_fast_{config,run}.m` (fast smoke, 0.67 h, 24/24 figures) and `scripts/paper/topk_med_{config,run}.m` (medium, run overnight 2026-09-12/13 into `data/topk_med` and `figs/topk_med`; TR “really likes” the output).

1.3 Medium-run headline numbers (figs/topk_med, n = 500, 3 seeds, K = 200, 40 s)

regime	n_+	h_{KS} (bit/s)	D_{KY}	λ_1
no adaptation, noise on	5 [4, 8]	12.2 [8.7, 20.9]	11.6 [9.6, 16.7]	+2.98
no adaptation, noise off	5 [4, 8]	10.3 [6.9, 18.3]	11.2	+2.79
single-timescale, noise on	7 [4, 8]	2.5	15.7	+0.48
single-timescale, noise off	8 [8, 10]	3.6	19.4	+0.70
multiple-timescale, both	0	0	0	-0.109

Transient amplification (J_{xx} , median over 300 states): $\omega = 82 / 74 / 46 \text{ s}^{-1}$, $\alpha = 7 / 6 / 0.3$ for no / single / multiple-timescale adaptation. Note noise RAISES h_{KS} in the no-adaptation regime here (opposite of Engelken et al.’s fluctuating-input result); with 3 seeds the ranges overlap — worth a look, not yet explained.

1.4 Notes written

`docs/notes/Lyapunov_estimation_methods.md` (§2.4 alignment/spread, §4.2 cost, §4.4 what the sweeps report), `Cov_Lya_vectors.md`, `topK_reading_list.md`, `CLV_and_spectrum_reading_list.md` (papers now in `C:\Users\m218089\Desktop\github_repos\PDF2md\StabilityPaper\*_figures\*.md`), **`Non_normal_amplification.md`** (+ PDFs). Manuscript: `C:\Users\m218089\Desktop\github_repos\StochasticPlasticDynamicalSystemPaper\Manuscript5` (TR will rewrite it around top-K; image links to `Fig_Sensitivity_mean_rate_medians` change case, param-space sheets are per measure). Kreiss constant code (Mitchell 2020/2021, wrapper `test_Kreiss/Kreiss.m`) lives in `C:\Users\m218089\Desktop\github_repos\Kreiss`; decided NOT to use it (see §3.3).

2. The non-normal amplification discussion (2026-09-13)

2.1 What the current figure shows and why J_{xx}

`fig_transient_amplification`: $\omega(J_{xx})$ is the largest instantaneous growth rate of $\|\delta x\|$ (max eigenvalue of the symmetric part), $\alpha(J_{xx})$ the asymptotic rate (max Re eigenvalue); $\omega - \alpha$ per state is the “non-normal margin”. $\omega \gg \alpha$ in every regime because a Dale’s-law W has a rank-one mean structure whose largest singular value ($\sqrt{n} \cdot \mu$) far exceeds its eigenvalue outliers; the direction achieving ω is the E-minus-I difference mode: Murphy & Miller’s balanced amplification, Hennequin’s non-normal amplification. Adaptation lowers ω $82 \rightarrow 46$ (operating point: STD depresses W_{eff} , SFA moves neurons to shallower ϕ').

Why the dendritic block and not the full J : the numerical abscissa is not invariant to a diagonal rescaling of the state, and the full J mixes rows in units $1/\tau_d = 10$, $1/\tau_a = 0.1\text{--}4$, $1/\tau_{\text{rec}} + r/\tau_{\text{rel}} \approx 0.3\text{--}5 \text{ s}^{-1}$ with asymmetric cross

couplings. TR’s point: in THIS model x , a , b are all dimensionless and $O(1)$, so the full- J ω is not meaningless (it came out 50-180, same order as J_{xx}) — it measures whole-state instantaneous growth in a stated norm, but is not invariant, so show it with the norm stated and do not build a claim on it. The deeper limitation of BOTH is that ω is instantaneous ($t = 0^+$): adaptation’s *dynamic* negative feedback (a , b responding over 0.25-10 s) is invisible to it. The J_{xx} figure therefore shows only the static (operating-point) part of adaptation’s effect and UNDERSTATES the reduction — conservative.

2.2 Agreed recommendations (TR agreed to all three)

1. **Transient gain, frozen vs active — the primary measure.** $G(t) = \|P_x e^{Jt} P_x^T\|_2$: perturb only dendritic states, let the whole system respond, measure only on dendritic states. Scale-invariant (x-in, x-out, one unit), finite-time (no stability requirement), contains adaptation’s dynamic feedback. Compute twice per sampled state: through J_{xx} (adaptation FROZEN = the conventional rate-network picture) and through the full J (adaptation ACTIVE); the gap between the two curves is adaptation’s dynamic contribution and the figure that says adaptation belongs in the conceptual model of connectivity and dynamics (a thesis of the paper). Report peak gain $G_{\max}$ and peak time t_{peak} per regime, frozen and active.
2. Keep the $J_{xx} \omega/\alpha$ figure as the “conventional view” (maps onto the literature; shows the operating-point effect).
3. Add the full- J ω as a supplementary panel with the norm stated.

Direction of the perturbation — no choice needed. $G(t)$ is the worst case over all dendritic directions; the optimal direction is the top right singular vector of the x-to-x propagator at that t (differs across t). Because propagating the n dendritic unit vectors gives the whole $n \times n$ propagator, three further readings come free: the **noise-average gain** (Frobenius/ $\sqrt{n}$ = mean-square amplification of isotropic perturbations = what the model’s own additive noise on x experiences), the gain along the **E/I difference mode** (Murphy-Miller’s prediction; compare with worst case), and along the **leading Lyapunov direction** (x part of $Q_{\text{final}}(:, 1)$; how much of the available amplification the dynamics actually recruit). Also report the optimal vector’s E/I structure and participation ratio at the peak.

Cost. Full J at $N = 4000$: propagate $n = 500$ columns with the matrix-free jacobian_times (~ 30 ms per call at $K = 500$) $\rightarrow$ about 1 min per state for a 2 s horizon at 400 Hz; 20 states per regime ≈ 1 h total. J_{xx} (500×500 dense): expm at a few t , trivial. Horizon: 2 s first (dendritic + fast STD timescales); the 10 s SFA rung needs a longer horizon on fewer states if the curves have not settled.

Caveat for all of it: tangent-space linearisation along a noisy trajectory $\rightarrow$ distributions over states; ignores saturation, so G is an upper envelope of real amplification.

2.3 Kreiss constant — considered, left out

$K(A) = \sup_{\{Re z > 0\}} Re z \| (zI - A)^{-1} \|$ is a certified LOWER bound on the peak transient gain ($K \leq \sup_t \|e^{At}\| \leq e^{N \cdot K}$). Left out because: it needs a stable matrix (J_{xx} has $\alpha = +7$, $+6$ in two regimes; only the multiple-timescale regime qualifies); it has the same state-scaling dependence as ω on the full J (an x-in/x-out K would need modifying Mitchell’s inner resolvent-norm evaluation); on the full J ($N = 4000$) Mitchell’s dense code is hours per state whereas $G(t)$ is a minute; and G is the quantity itself, not a bound. Could return as a cross-check on J_{xx} in the stable regime ($K \leq G_{\max}$; compare Kreiss’s optimal direction with G ’s top singular vector).

3. Where we are, and what to add next (agreed ordering)

What we have. Jacobian side: ω , α of J_{xx} along the trajectory (static). Trajectory side, per sweep job: fraction of time the local Lyapunov rate is positive, p95 of the 0.2 s finite-time exponent, mean positive-excursion length, leading-vector block fractions. Spectrum side: n_+ , h_{KS} , D_{KY} per regime. Missing: the DYNAMIC picture — what a transient does once adaptation responds, and whether the network’s own transients are the ones non-normality predicts.

Next 1 — Transient-gain stage (build first). New stage `run_transient_gain(model on run_numerics_verification / run_lyapunov_spectrum: name-value cfg, cost table with :badMode, P struct, ensure_pool, results/cond_names/condition_t, schema, register in run_all_paper_analyses, test_run_modes, the configs) + fig_transient_gain`. Per regime, ~ 20 states from the same seeds as the eig stage (`build_from_preset, rng_seeds [1 2]`, sample after the warm-up), horizon 2 s: $G_{\text{worst}}(t)$, $G_{\text{noise}}(t)$, $G_{EI}(t)$, $G_{Lyap}(t)$, frozen (J_{xx} via expm) and active (full J via jacobian_times on the n -column basis [`zeros; I_n`] in the x rows, Heun as in `lyapunov_topk`, top singular value of the x rows

every ~ 0.02 s), optimal-vector E/I structure at the peak. Outputs: $G_{\max}$, t_{peak} , per regime $\times$ frozen/active; figure: three panels (regimes), two curves each with seed/state bands, plus a table.

Next 2 — Intermittency: are the excursions recruited transients? Same machinery, different sampling: in the single-timescale regime, sample states at local-rate excursion ONSETS and in QUIET stretches (the stored `local_rate_lead / t_lya_lead` mark them). Ask: is the optimal transient direction at onset states aligned with the leading Lyapunov direction, and is $G(t)$ at onset states larger than at quiet ones? If yes, the intermittent divergence is non-normal amplification triggered by the network’s own fluctuations — a mechanistic account of the Introduction’s “transient divergence” claim. Fold into the same stage as a sampling rule.

Next 3 — Stimulation that engages adaptation without a large transient (TR’s idea). The manuscript’s stimulation section engages adaptation crudely (DC) on a limit-cycle bursting network. TR’s proposal: on a network that is asymptotically stable but has positive-LLE transients (intermittent), find stimulation directions that are LEAST aligned with transient amplification in the short run (avoid an immediate large transient) yet strongly engage adaptation, so that adaptation reduces transient amplification later in time. Linear formulation: for an input direction v on x , short-time cost = $\int_0^\tau \|P_x e^{Jt} P_x^T v\|^2 dt$ ($\tau \approx$ a few hundred ms), long-time benefit = $\|P_a e^{Jt} P_x^T v\|$ at t of a few seconds (engagement of SFA/STD states); both are quadratic forms in v , so the best direction is a generalised eigenvector of two $n \times n$ matrices built from the same propagators as Next 1. Prior: the optimum is near the E/I SUM mode (the difference mode is what amplifies; a uniform push engages every neuron’s adaptation) — which would put the manuscript’s DC stimulation on principled ground, or reveal a better direction. Then a NONLINEAR check: stimulate the actual model along v vs along the worst-case direction with a matched pulse and measure the subsequent ω , the local-rate excursion rate and discharge rate. Separate stage; likely a separate section or follow-on paper.

Prerequisite for 2 and 3 — the right network. `bursting_pairs` is a limit cycle, not an intermittent stable system. The medium sweep (`data/topk_med/1D_sensitivity_*`, `param_space_*`) stores per job λ_1 , `frac_local_positive`, `p95_finite_0p2s`, `mean_positive_excursion_s`: QUERY it for grid points with $\lambda_1 < 0$, a high positive-time fraction and long excursions (load with `ParamSpaceAnalysis2.from_dir`, filter results). Do this before designing Next 3.

4. If resuming after a compaction: read these first

- This file; then commit messages `git log 5a9d409..3a64568`.
- `docs/notes/Non_normal_amplification.md` (§3 J_{xx} rationale, §5 the four quantification options; item 3 there = Next 1 above).
- `src/model/lyapunov/lyapunov_topk.m` (Heun loop with `opts.jac_times`; copy its propagation for the transient-gain stage), `SRNNCellTypePairs.jacobian_times` (~line 2141), `lya_summary`, `make_state_layout` (`layout.x` = last n indices; `.a{q}`, `.b{pre,post}`).
- `src/analysis/run_eig_heatmap.m` (`sample_eigenvalues`: how states are sampled, $J_{xx} = J(\text{end-n+1:end}, \text{end-n+1:end})$), `run_lyapunov_spectrum.m` (stage template), `src/figures/fig_transient_amplification.m`.
- `scripts/paper/run_all_paper_analyses.m` stage table (~line 135), `scripts/tests/test_run_modes.m` (~line 69), `scripts/paper/topk_med_config.m` figure registry (`add(F, name, fn, in_paper, args)`).
- Figure contract: arguments with `data_file/out_dir/save/visible/run_dir/preset_name`; `setup_paths`; `default_out_dir`; `manuscript_style`; `resolve_data_file`; return `struct('figs', fig, 'files', {}, 'source', data_file)` and `save_figure_stable + existing_outputs`.
- House rules: MATLAB only via MCP (long runs: launch from the MATLAB prompt or capture with `evalc` to a log — the MCP client aborts after 30 min of silence); clear classes after `classdef` edits (say so); stage by name; **no Co-Authored-By trailer in commit messages** (CLAUDE.md, updated); `figs/` and `data/` gitignored; note PDFs are force-added.

5. Addendum 2026-09-13 (after compaction): Next 1 and Next 2 built

Same session (90c5825b-...), same machine; commits 478052a, 71bf5a5 and the docs commit after them. Working tree clean at the end.

Built. `SRNNCellTypePairs.transient_gain/leading_direction_at/excursion_samples(+test_transient_gain)`; stage `src/analysis/run_transient_gain.m` (registered after `lyapunov_spectrum` in `run_all_paper_analyses`, `test_run_modes`, and the figure registries of `paper_config`, `single_multi_TS_independent_config`, `med_config`,

topk_smoke_fast_config, topk_med_config); figures fig_transient_gain (Fig_Transient_Gain) and fig_transient_gain_excursions (Fig_Transient_Gain_Excursions), each with a _table.md; test_transient_gain_stage; scripts/examples/find_intermittent_stable.m; CLAUDE.md bullet; Non_normal_amplification.md §6 (the results, read that first).

Smoke (data/transient_gain_smoke, figs/transient_gain_smoke, fast, $n = 500$, 1 seed, 11 min): active 1-s $G_{\max} 1\ 200 / 30 / 8.6$ for no / single / multiple-timescale adaptation; freezing the full J instead of J_{xx} alone cuts the frozen gain $\sim 80\times / \sim 6\times$ in the adapted regimes (the dynamic feedback). The frozen gain in a regime with $\alpha(J_{xx}) > 0$ grows without bound, so its “peak” is the horizon – compare at fixed t . The active gain is NOT bounded by the frozen ones (above them on the small test nets and in the stable regime at $n = 500$). The worst-case direction has $|\cos| \leq 0.15$ with the leading Lyapunov direction; the E/I difference mode is the best named direction, the E/I sum mode the worst (< 1 when adapted); the noise-average gain is ~ 1 in the adapted regimes.

Next 2 status. The sampling works but 20 s gives 0-2 onsets/quiets per regime; the medium mode (T 40 s, 2 seeds, up to 8 + 8) is the first real contrast. topk_med_config now includes the stage: rerunning topk_med_run (delete data/topk_med first, or run the stage alone into data/topk_med/transient_gain with run_transient_gain('run_mode', 'medium', 'out_dir', ...) – about 1 h) gives it.

Next 3 prerequisite done. find_intermittent_stable() over data/topk_med: 78 / 78 / 114 intermittent-but-stable jobs per regime; best candidates the multiple-timescale regime at $\mu_{IE_relative} 2.06$ or $\mu_{EE_relative} 13.4-15.7$ (1-D sensitivity, several reps, $\lambda_1 \approx -0.01$ to -0.1 , p95 finite-time exponent 10-13 s^{-1}), and single-timescale at $n = 190$.

Not done / raise only. Next 3 itself (the generalised eigenproblem for stimulation directions and the nonlinear check); Kreiss; CLVs; a noise-off replicate of the transient-gain stage; a longer horizon for the 10-s SFA rung. Also unexplained from §1.3: noise RAISES h_{KS} in the no-adaptation regime.